

KEM, Russian Federation

WMO: 225220

Lat: 64.950N

Lon: 34.650E

Elev: 8

StdP: 101.23

Time Zone: 3.00 (E03)

Period: 90-06

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.1	-24.0	-29.6	0.2	-26.9	-26.2	0.3	-23.7	12.2	-3.4	11.0	-3.8	2.1	290	0.773

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.7	23.6	16.0	21.2	15.3	19.3	14.7	17.3	21.5	16.2	19.9	15.2	18.5	4.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.6	11.1	19.1	14.4	10.3	17.8	13.5	9.7	16.8	48.8	21.7	45.4	20.1	42.5	18.6	21.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.6	9.4	8.4	-29.4	27.1	5.0	0.8	-33.0	27.6	-35.9	28.1	-38.7	28.5	-42.3	29.1		
(5)			-29.6	19.6	4.9	1.2	-33.1	20.5	-36.0	21.2	-38.7	21.8	-42.3	22.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.5	-9.8	-9.1	-5.2	-0.6	4.4	11.0	13.7	13.0	8.1	2.1	-4.1	-6.3		
	DBStd	9.57	7.14	7.59	5.03	4.77	3.94	4.31	3.16	3.13	3.42	4.45	5.23	5.30		
	HDD10.0	3414	613	535	471	317	182	38	4	6	75	244	424	504		
	HDD18.3	6153	872	768	729	567	433	225	148	167	306	502	674	763		
	CDD10.0	314	0	0	0	0	7	68	119	101	19	1	0	0		
	CDD18.3	11	0	0	0	0	0	4	4	2	0	0	0	0		
	CDH23.3	61	0	0	0	0	1	27	12	21	1	0	0	0		
	CDH26.7	4	0	0	0	0	0	3	0	2	0	0	0	0		
	WSAvg	4.3	4.8	4.4	4.5	4.0	4.0	4.0	4.0	4.0	4.2	4.7	4.7	4.8		
	PrecAvg	434	20	15	16	24	40	56	46	60	54	49	37	26		
(15) Precipitation	PrecMax	708	42	35	36	47	97	122	110	113	109	84	59	69		
	PrecMin	274	6	3	3	1	7	10	2	15	18	6	10	2		
	PrecStd	88	10	9	8	12	22	32	27	22	30	20	13	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.4	4.2	5.4	13.6	19.9	27.2	25.2	26.1	22.1	12.4	6.1	3.6		
		MCWB	1.6	2.2	2.4	7.7	12.6	17.1	17.1	17.4	15.9	10.3	4.8	2.0		
	2%	DB	1.8	2.9	4.1	9.9	16.1	24.0	22.5	22.7	17.4	10.7	4.6	2.4		
		MCWB	0.7	1.8	1.7	5.5	10.2	15.6	15.9	16.1	13.1	9.1	3.9	1.3		
	5%	DB	0.7	1.3	2.8	7.5	13.0	20.7	20.5	20.2	14.7	9.6	3.6	1.5		
		MCWB	-0.2	0.4	0.9	4.3	8.5	15.0	15.3	15.0	12.4	8.2	2.9	0.6		
	10%	DB	-0.5	0.2	1.4	5.5	10.4	18.1	18.7	18.2	12.7	7.9	2.5	0.6		
		MCWB	-1.3	-0.6	0.0	3.2	7.0	13.9	14.8	14.4	10.9	6.6	1.8	-0.1		
	0.4%	WB	1.9	2.7	3.2	8.2	13.7	19.1	18.2	17.9	15.7	11.1	5.2	2.2		
		MCDB	3.1	3.5	4.6	13.0	17.7	23.9	22.4	23.7	20.4	12.2	5.7	2.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	0.9	1.7	2.0	5.9	11.1	17.6	17.0	16.9	14.0	9.6	4.0	1.5		
		MCDB	1.9	2.7	3.3	9.0	15.6	21.4	20.7	21.5	16.6	10.4	4.4	2.2		
	5%	WB	0.0	0.6	1.0	4.6	8.9	15.5	16.0	15.7	12.5	8.1	3.0	0.8		
		MCDB	0.7	1.2	2.4	7.1	12.6	20.3	19.2	19.1	14.3	9.3	3.7	1.4		
	10%	WB	-1.2	-0.6	0.1	3.3	7.3	13.6	15.1	14.8	11.2	6.8	1.9	-0.1		
		MCDB	-0.5	0.2	1.4	5.4	10.2	18.1	18.2	17.7	12.5	7.9	2.4	0.5		
	5% DB	MDBR	6.3	6.2	6.6	6.5	5.9	6.4	5.7	6.1	4.8	4.2	4.8	5.5		
		MCDBR	6.2	4.9	6.6	8.3	11.5	11.0	9.1	10.4	8.2	5.5	4.1	4.4		
		MCWBR	5.7	4.1	5.0	5.1	6.5	6.0	4.4	5.2	4.9	4.6	3.6	4.1		
		MCWBR	6.2	4.7	6.0	7.7	10.7	10.0	7.8	8.8	7.1	5.2	4.1	4.4		
(35) Mean Daily Temperature Range	5% WB	MCWBR	5.7	4.0	5.0	5.2	6.6	6.2	4.4	4.8	4.5	4.7	3.6	4.2		
	taub		0.218	0.280	0.309	0.342	0.354	0.350	0.378	0.355	0.331	0.303	0.346	0.358		
		taud	1.953	2.093	2.152	2.273	2.425	2.483	2.432	2.472	2.518	2.364	2.563	2.613		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		283	591	755	824	854	866	824	811	756	588	275	118		
		Edh at Noon	24	61	90	102	97	94	97	85	65	48	21	12		
	RadAvg		0.12	0.70	2.24	3.74	4.66	5.46	5.01	3.57	2.02	0.73	0.16	0.04		
(45) All-Sky Solar Radiation	RadStd		0.01	0.11	0.23	0.36	0.41	0.48	0.55	0.37	0.21	0.10	0.02	0.00		

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	+0.64	N/A	N/A	N/A	N/A	N/A	N/A	-227	N/A

Nomenclature: See separate page